
BIOLOGY AND ENVIRONMENTAL SCIENCES

Studies on Diatom Flora of Schirmacher Oasis, East Antarctica

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ABSTRACT

The Diatom flora of Schirmacher Oasis, East Antarctica was investigated during XXIIIrd (2003-2004) Indian Scientific Expedition to Antarctica. An assessment of the diversity of diatom flora of Schirmacher Oasis resulted in a total of 18 species, representing 11 genera of which *Achnanthes* (2), *Fragilaria* (3), *Synedra* (1), *Luticola* (2), *Navicula* (3), *Stauroneis* (1), *Diadismis* (1), *Pinnularia* (1), *Gamphomema* (1), *Hantzschia* (1) and *Nitzschia* (2). Of 18 taxa found, 8 were first time for E. Antarctica. This paper deals the taxonomical description of all diatom. This study reveals that the species composition of diatoms are more in the Western part than the Eastern part of the Schirmacher Oasis. *Pinnularia*, *Navicula*, *Luticola*, *Nitzschia* and *Hantzschia* were abundant in Schirmacher Oasis.

INTRODUCTION

The first systematic exploration of the Polar water of Antarctica was conducted by Englishman James Cook who circumnavigated Antarctica twice between 1772 and 1775. After that a number of investigations were made to explore diversity of marine coastal vertebrates, phytoplankton, and several workers studied distribution and taxonomical evaluation of algal flora of different places of Antarctic ice continent by West and West (1911), Fritsch, F.E. (1912), Komarek and Ruzicka (1966), Parker *et al.* (1972), Hirano (1979), Broady (1982), Howard Williams *et al.* (1986) and Hawes (1988). Since then about 700 algal taxa have been reported from the Antarctic continent by Pankow *et al.* 1991. A detailed study of diversity, distribution and dispersal of Antarctic terrestrial algae was given by Broady (1996). The investigation on diatom flora also made from various part of Antarctica by Fukushima (1962), Ohtani (1986), Andreoli *et al.* (1995), Jones (1996), Kawecka and Olech, (1993) and Temniskova-Topalova *et al.* (1999).

Schirmacher Oasis in particular has also attracted considerable attention from the point of view of algal studies (Aleshinskaya and Bardin, 1965; Komarek and Ruzicka, 1966; Simonov, 1971; and Pankow et al., 1987). Altogether some 63 taxa were reported. However, Kashyap, 1988; Pandey et al., 1995; Pandey and Kashyap, 1995; Gupta & Kashyap, 1998; Shukla et al., 1997, 1999 were studied the Algal flora of Schirmacher Oasis. Most of the work is mainly confined to morphology, floristic, physiology, nutrient status, life cycle and diversity of Blue green algae only. Gupta (2002) reported 10 diatom species from Schirmacher Oasis. Attention has paid to Bacillariophyceae, Chlorophyceae, Chrysophyceae, Dinophyceae and Dinophyceae were very meager. It is thus apparent that a considerable lacuna in the study of algal diversity of Schirmacher Oasis in general, except for the Cyanophycean elements still persist from Indian point of view. It was felt that a comprehensive but concise compilation is a requirement. The present program therefore is an attempt to bring together the basic way of identification and documentation of all diatoms belonging to many of the genera and species occurred abundantly and commonly in fresh water ecosystems of Schirmacher Oasis, East Antarctica.

Diatoms are single celled algae of the class Bacillariophyceae. The skeleton of a diatom or frustule is made of very pure silica. This skeleton is divided in to two parts, one of which (epitheca) overlaps the other (hypotheca) like the lid of box or petridish. Diatoms extremely abundant in aquatic ecosystems, it is estimated the 20% to 25% of all organic carbon fixed on the planet is carried out by diatoms (Werner, 1977). Diatoms are thus a major food resource for aquatic microorganisms and animal larvae, and are a major source of atmospheric oxygen. Diatoms are also used for water quality evaluation (Almedia, 1998). The silicified cell wall gives a beautiful look. Diatomaceous earth, a substance composed of fossil diatoms, is used for making filters, insulation, abrasives, paints, etc.

STUDY AREA

Schirmacher Oasis (70° 44' S to 11° 54' E) is a group of low lying – hills, essentially snow and ice-free high polar rock desert in the Eastern Droning Maud Land, East Antarctica (Fig. 1). It is 70 km away from Prince Astrid Coast. The Oasis oriented in east – west direction, lies between two types of large ice bodies viz., glacier in the south and ice shelf at the north of the Oasis. It covers an area of about 34 km². The altitudes lie between the local zero level and 228 m with an average of 100 m. During the polar summer the ice melts and water flows in the form of streams often adding

water to the lakes. The climate is relatively mild in the low altitude, with air temperature over the glacier ice between -4.5 and $+10.5$ °C during mid summer (December 2003 – January 2004) when melted water is abundant.

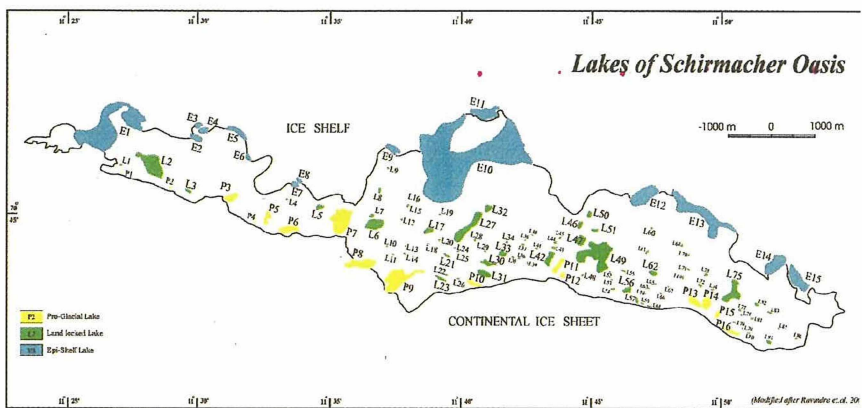


Fig. 1: Map of Schirmacher Oasis

MATERIALS AND METHODS

Field work was carried out during the XXIIIrd Indian Scientific Expedition to Antarctica in the year 2003–2004. Algae growths are abundant and readily visible on the surface of the rocks, boulders, weathered soils and moss carpets. The author collected pieces of moss, upper layer of bottom sediments were removed using spoon and water samples from different lakes of Schirmacher Oasis, East Antarctica (Plates 1 & 2) to study the diatom flora. The collected samples were fixed with 4% formalin solution and brought to Botanical Survey of India, Port Blair for further study. Part of the samples was oxidized in a mixture of sulphuric acid and potassium dichromate (volume 3:1) to remove the organic matter. Then the material was centrifuged and digested again. It was then rinsed with distilled water several times, to get rid of acid. The cleaned samples were examined under light microscope (Olympus CX 41). Taxa were identified using recent references (Desikachary, 1987; Prasad & Srivastava, 1992; Anand, 1998; Kawecka et al., 1998; and Antarctic diatom web: <http://www-personal.umich.edu>). All collected samples have been deposited in Botanical Survey of India, Port Blair for further reference.

**PLATE 1—SAMPLED AREA IN SCHIRMACHER OASIS,
EAST ANTARCTICA**



*A view of land lock lake at West
S.Oasis*



Water fall in North S.Oasis



Lush Algal growth



A portion of long lake



Pro-glacial lake at Western S.Oasis

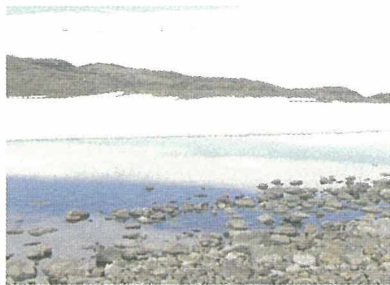


Ice melting in mid summer

**PLATE 2—SAMPLED AREAS IN SCHIRMACHER OASIS,
EAST ANTARCTICA**



*Ice free land lock lake at North
S.Oasis*



A view of stream in mid summer



Lake near German camp



*A panoramic view of lake at
D.G.snout*



Algal mats



Salt deposit in land lock lake

TAXONOMICAL COMMENTS

Preliminary data show the occurrence of 18 taxa of diatoms from different environments of Schirmacher Oasis, East Antarctica.

Class : Bacillariophyceae

Order : Bacillariales

Family : Achnanthaceae

1. *Achnanthes exigua* Grun.

Valves smaller, linear, elliptical with somewhat parallel margins and deeply constricted, slightly produced, rostrate ends; hypovalves with thin, straight, median raphe; axial area narrow, linear, gradually widening towards center; central area broad, stauros reaching the sides, striae fine dense, lineate, delicate slightly radial throughout the valve; epivalve with median, narrow pseudo-raphe; striae fine, lineate, radial and parallel throughout the valve.

Cell range : Length 15 μm , width 4-6 μm and striae, 29-31 in 10 μm .

Distribution : scattered in Schirmacher Oasis.

In India : Andaman; Andhra Pradesh; Gujarat; Karnataka; Kerala; Maharastra; Tamil Nadu and West Bengal

2. *Achnanthes minutissima* (Kuetz.) Grun.

Valves smaller, linear; narrow, linear lanceolate with rounded ends; thin raphe; axial area narrow; central area slightly wider.

Cell range : Length 18.5-31.5 μm ; width 3.5-4 μm and striae, 25 in 10 μm

Distribution : Scattered in Schirmacher Oasis.

In India : Maharastra

Family : Bacillariaceae

3. *Hantzschia amphioxys* (Ehr.) Gurn.

Valves freely arcuate, linear dorsal side convex, ventral side somewhat concave with distinct depression in the middle; ends broadly produced and bluntly rounded; keel punctae coarse distinct, slightly elongated, median two some what placed distantly, rudiments of central nodules present below the depression of ventral valve; striae fine, lineate parallel throughout the valve.

Cell range : Length 53.5-75.5 μm ; width 8-10.5 μm ; striae, 18-21 in 10 μm and punctae, 8-10 μm in 10 μm .

Distribution : Common in Schirmacher Oasis.

In India : Andaman; Andhra Pradesh; Gujarat; Karnataka; Maharashtra; South India; West Bengal.

4. *Nitzschia obtuse* W.Simth.

Valves long, linear with almost parallel margins and having slight constriction in the middle, towards apices attenuated unilaterally while other valve slightly curved to give the valve somewhat sigmoid appearance having rounded ends; keel punctae coarse, excentric, parallel to the valve, median two placed distantly, giving the impression of the presence of central nodule; striae fine, lineate, delicate, parallel throughout the valve.

Cell range : Length 93.5-145.5 μm ; width 7-8.5 μm ; striae, 28-30 in 10 μm and punctae, 8-9 in 10 μm .

Distribution : Common in Schirmacher Oasis.

In India : Andaman; Andhra Pradesh; Maharashtra; Uttar Pradesh; West Bengal.

5. *Nitzschia capitellata* Hust.

Valves long, linear with parallel margins and broadly cuneate slightly constricted rounded – capitate ends; carinal dots coarse, very slightly elongated, median two some what distantly placed; striae very fine, lineate, delicate, parallel throughout the valve.

Cell range : Length 67.5 μm ; width 6.2 μm ; striae, 30-32 in 10 μm .

Distribution : Scattered in Schirmacher Oasis.

In India : Andaman

Family : Fragilariaceae

6. *Fragilaria intermedia* Grun.

Valves linear with parallel margins; gradually tapering towards ends and rounded slightly capitate; axial area narrow, linear.

Cell range : Length 55- 69.5 μm ; width 6-7.25 μm ; striae, 2-13 in 10 μm .

Distribution : Scattered in Schirmacher Oasis.

In India : Maharashtra.

7. *Fragilaria intermedia* Grun. var. *robusta* Venks.

Valves linear with parallel sides; gradually tapering towards ends, slightly capitate ends; striae coarse and distinct and one side absent in the middle area and therefore with a unilateral central portion.

Cell range : Length 87.5-114.5µm; width 10-12.5 µm; striae, 10-13 in 10 µm.

Distribution : Scattered in Schirmacher Oasis and it is addition to this Oasis.

In India : Uttar Pradesh.

8. *Fragilaria virescens* Ralfs.

Valves long, linear with parallel margins, attenuated rostrate ends; pseudo-raphe present; axial area narrow, linear; striae fine, lineate, parallel throughout the valve.

Cell range : Length 70.25-77.5µm; width 5.5-6.2 µm; striae, 12-17 in 10 µm.

Distribution : Scattered in Schirmacher Oasis.

In India : Andaman; Eastern Himalaya; Maharashtra.

9. *Synedra ulna* (Nitzsch.) Ehrenb.

Valves solitary, linear much elongated, gradually attenuated to the rostrate ends; pseudo-raphe distinct, narrow, widening towards center; central area quadrangular having small lineate, transverse, parallel throughout the valve.

Cell range : Length, 110 - 135.5µm; width, 6.2-7 µm; striae, 11 -13 in 10 µm.

Distribution : Rare in Schirmacher Oasis and new to the Oasis.

In India : Andaman; Andhra-Pradesh; Assam; Bihar; Gujarat; Jammu and Kashmir; Karnataka; Maharashtra; Rajasthan; South India, Uttar-Pradesh; West Bengal.

Family : Gomphonemaceae

10. *Gomphonema lanceolatum* Ehren.

Valves linear to lanceolate with attenuated rounded ends; raphae thin, straight, median, central noudules bent unilaterally, terminal fissures curved forming question mark; axial area narrow, linear- lanceolate, central area broad, somewhat rectangular, unilateral, having an isolated puncta on opposite side, striae coarsely punctuate, radiate and parallel, median short striae widely placed.

Cells range : Length, 51.5-53.5µm; width, 8.5-11.4 µm; striae, 10-12 in 10 µm.

Distribution : Scattered in Schirmacher Oasis.

In India : Andaman; Gujarat; Karnataka; Maharashtra; Madhya Pradesh; Tamil Nadu; Uttar Pradesh.

Family : Naviculaceae

11. *Diadesmis contenta* (Grun.ex Van Heurck) D.G.Mann

Syn. : *Navicula contenta* Grun.ex Van Heurck.

Valve linear, slightly gibbous in the middle portion; poles somewhat swollen and rounded. Axial area small or indistinct. Central area wide, generally rounded, often with a thickening of the valve beneath the central area; striae linear; generally only one row of striae on each half of the valve face.

Cells range : Length, 6-20 µm; width, 2-3 µm; striae, 28-33 in 10 µm.

Distribution : Common in Schirmacher Oasis and an addition to the Oasis.

12. *Luticola muticopsis* (Van Huerck) D.G.Mann

Syn. : *Navicula muticopsis* Van Huerck

Valve sub-elliptical, with lateral margins extended or sub-parallel; endings rostrate to slightly capitate; Axial area present, widening towards the central region of the valve; central areatypically elliptical or oval shaped; raphe straight; striae radiate throughout.

Cells range : Length, 17-26 µm; width, 6-8 µm; striae, 15 -18 in 10 µm.

Distribution : Scattered in Schirmacher Oasis.

13. *Luticola muticopsis* f. *capitata* (Carlson) D.G.Mann

Valve distinctly capitate; possibly with one lateral margin slightly convex and the other parallel. Axial area present, widening towards the central region of the valve. Central area an acute- angled sub- fascia, usually with a single row of poroids around the outer edge. A solitary stigma present on the sub-fascia. Raphe straight; median and terminal raphe endings slightly hooked in the opposite direction of the stigma. Striae radiate throughout, consisting of 3-5 distinct puncta except in the median portion of the valve and at the poles.

Cells range : Length, 15 - 25 μm ; width, 8-11 μm ; striae, 14 -18 in 10 μm .

Distribution : Scattered in Schirmacher Oasis and an addition to the Oasis.

14. *Navicula cuspidate* var *ambigua* (Ehren.) Cleve

Valve long, broadly lanceolate with narrowly constricted much produced, flatly rostrate capitate ends; raphe thin, straight, median with distinct, unilaterally bent central nodules and curved terminal fissures; axial area narrow, linear; central area moderately wide and longitudinally elongated; valve surface striated, transverse striae lineate, parallel interrupted by longitudinal striae.

Cells range: Length, 67.5-78.5 μm ; width, 18.5-20 μm ; transverse and longitudinal striae, 19 -22 in 10 μm .

Distribution: Rare in Schirmacher Oasis.

In India : Andaman; Andhra Pradesh; Bihar; Gujarat; Karnataka; Maharashtra; Rajasthan; Tamil Nadu; Uttar Pradesh; West Bengal.

15. *Navicula cryptocephala* Kuetz.

Valves linear-lanceolate with constricted produced somewhat capitate ends; raphe thin, straight median with distinct central nodules; axial area narrow, linear surrounding raphe and central nodules; central area somewhat rounded; striae lineate, short in central area, radiate in the middle becoming convergent towards apices.

DIATOMS FROM SCHIRMACHER OASIS, EAST ANTARCTICA



Achnanthes exigua



Achnanthes. minutissima.



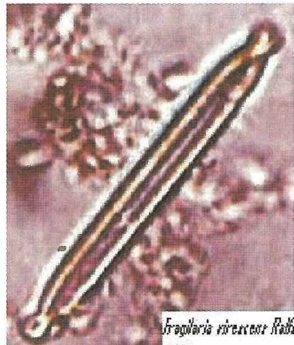
Diadesmis contenta



Fragilaria intermedia



F. intermedia var. *robusta*



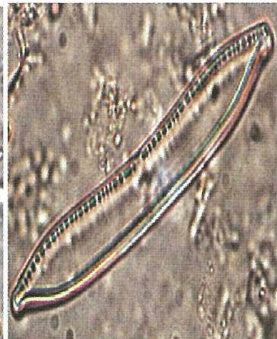
F. virescens



Synedra ulna

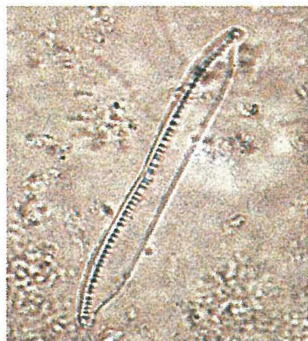
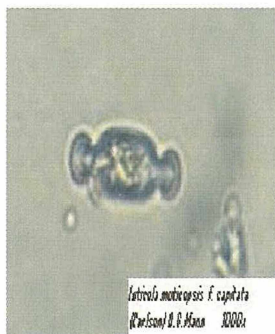
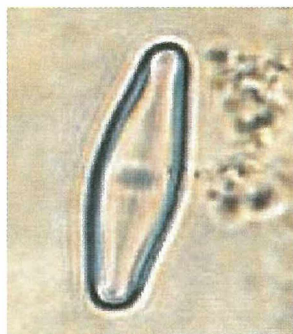
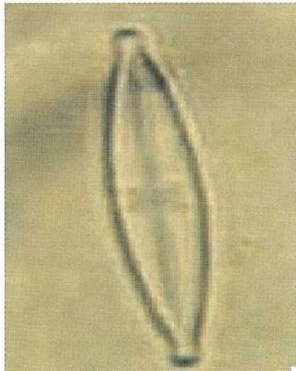


Gomphonema lanceolatum



Hantzschia ampioxys

DIATOMS FROM SCHIRMACHER OASIS, EAST ANTARCTICA

*Nitzschia obtuse**Nitzschia capitellata**Luticola muticopsis**L. muticopsis f. capitata**Navicula radicata**Navicula cryptocephala**Navicula cuspidate**Pinnularia borealis**Stauroneis anceps*

Cells range : Length, 30.2-32 μm ; width, 5.9-6.6 μm ; striae, 19 -22 in 10 μm .

Distribution : Rare in Schirmacher Oasis.

In India : Andaman; Andhra Pradesh; Eastern Himalaya; Gujarat; Karnataka; Maharashtra; Rajasthan; Uttar Pradesh; West Bengal.

16. *Navicula radios* Kuetz.

Valves linear- lanceolate, gradually attenuated towards rounded ends; raphe thin straight, median with distinct closely set, unilaterally bent central nodules; axial area linear; central area broad, some what elliptical; striae coarse lineate, curved and slightly radiate in the middle becoming convergent towards the poles.

Cells range : Length, 59-65 μm ; width, 9-12 μm ; striae, 9-11 in 10 μm .

Distribution : Scatted in Schirmacher Oasis.

In India : Andaman; Andhra Pradesh; Bihar; Karnataka; Maharastra; Rajasthan; South India.

17. *Pinnularia borealis* Ehrenb.

Valve broadly linear with rounded ends; axial area distinct; central area transverse and rounded; typically with one or two striae surrouding the sub fascia; raphe curved, median raphe endings bent to one side, distal endings sickle-shaped; thick striae, usually parallel, convergent at ends and radiate at the center.

Cells range : Length, 28-42 μm ; width, 8-10 μm ; striae, 5-7 in 10 μm .

Distribution : Common in Schirmacher Oasis.

In India : Cosmopolitan.

18. *Stauroneis anceps* Ehrenb.

Valve elliptical-lanceolate with protracted, capitate ends; axial area linear; central area a broad stauros; raphe simple and straight; slightly narrowed at the proximal and distal ends; striae radiate throughout.

Cells range : Length, 28-62 μm ; width, 8-12 μm ; striae, 20-25 in 10 μm .

Distribution : Rare in Schirmacher Oasis and an addition to the Oasis.

In India : Maharashtra and Uttra Pradesh.

Table 1 : Diversity of Diatoms in Schirmacher Oasis, Antarctica

Lake name*	Total no of lakes	No of samples collected	Genera identified
Land locked lake	33	77	<i>Achnanthes minutissima</i> <i>Achnanthes exigua</i> * <i>Fragilaria intermedia</i> <i>Fragilaria intermedia</i> . var. <i>robusta</i> * <i>Fragilaria virescens</i> <i>Synedra ulna</i> ** <i>Gomphonema lanceolatum</i> <i>Diadesmis contenta</i> ** <i>Navicula cryptocephala</i> <i>Navicula radiosa</i> <i>Navicula cuspidate var ambigua</i> * <i>Luticola muticopsis</i> <i>Pinnularia boralis</i> <i>Stauroneis anceps</i> ** <i>Hantzschia amphyoxis</i> <i>Nitzschia obtuse</i> <i>Nitzschia capitellata</i> *
Pro-Glacial lake	10	18	<i>Achnanthes minutissima</i> <i>Fragilaria intermedia</i> <i>Luticola muticopsis</i> f. <i>capitata</i> <i>Navicula radiosa</i> <i>Pinnularia boralis</i> <i>Hantzschia amphyoxis</i> <i>Nitzschia obtusa</i>
Epi-Shelf lake	11	17	<i>Fragilaria intermedia</i> <i>Luticola muticopsis</i> <i>Navicula radiosa</i> <i>Pinnularia boralis</i> <i>Hantzschia amphyoxis</i> <i>Nitzschia obtusa</i>

** New Genera recorded * Species reported

The 18 species of diatoms recorded from the various ecological niches of Schirmacher Oasis are presented in Table 1. Of 18 taxa, 8 species, namely *Achnanthes exigua* Grun., *Fragilaria intermedia* Grun. var. *robusta* Venks, *Synedra ulna* (Nitzsch) Ehrenberg, *Diadesmis contenta* (Gurn. Ex Heurck.) D.G.Mann, *Navicula cuspidate* var *ambigua* (Ehren.) Cleve, *Stauroneis anceps* Ehrenb. *Luticola muticopsis* f. *capitata* were reported for the first time from Schirmacher Oasis. Among the 11 genera the percentage of occurrence of each genus was as follows, Navicula-22%, Pinnularia-19%, Hantzschia- 15%, Achnanthes-11%, Fragilaria-9%, Nitzschia-9%, Gamphonema- 5%, Diadesmis-4%, Stauroneis-4% and Synedra-2%.

DISCUSSION

The development of microalgae is generally triggered by the light, temperature and nutrients. During the study period physicochemical properties were recorded from the sampled areas and the data are given in the Table 2. The atmospheric temperature was -4.5°C to +10.5°C and the water temperature was -1°C to + 9.5 °C. The lake water salinity ranging from 0.25 to 24.2ppt (this 24.2ppt recorded in land locked lakes only) and the pH value from 6.1 to 9.5. Due to higher salt concentration the algal diversity is more in land lock lakes compared to the proglacial and periglacial lakes. The current study reveled that the cosmopolitan species i.e *Pinnularia borealis*, *Hantzschia amphioxys*, *Nitschia obtuse* and *Luticola muticopsis* are presented in this Oasis but distribution of diatoms are differing from varies ecological factors. The western part of the land lock lakes harboring more variety of diatoms than the other part of the Oasis.

**Table 2 : Physicochemical parameters
(Between 01st Jan to 26th Feb)**

Sr. No	Parameters	Min.	Max.
1	Atmospheric temp. (°C)	-4.8	9.0
2	Water temp. (°C)	-1.0	9.6
3	Salinity(ppt)	0.26	24.5
4	pH	6.1	9.5

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